THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS V — MATHEMATICS · MODEL PAPER 07 ANSWER KEY

Answers and marking points | Maximum Marks: 80

UNDERSTANDING SHAPE AND SPACE — Marks 21

Q.	Marks	Answer / marking points
1	1	The side of each small square is 1 cm. The perimeter of the shaded figure is **(a) 14 cm**
2	6	**(i)** Name the type of each angle shown: (a) ∠A (b) ∠B (c) ∠C [3] (a) right angle (b) obtuse angle (between 90° and 180°) (c) reflex angle (more than 180°) **(ii)** (a) The hour hand moves from 12 to 4. Through how many degre… [3] (a) 4 × 30 = **120°** (b) a right angle (90°) (c) 180°
3	6	**(i)** Name the solid shape: (a) a die used in Ludo (b) an ice-crea… [3] (a) cube (b) cone (c) cylinder **(ii)** (a) How many vertices does a cube have? (b) How many edges d… [3] (a) 8 (b) 6 (c) 2 (two circular faces; the third face is curved)
4	8	Look at the plan of Raju's house in Bharatpur. **(a)** Find the area of the Drawing Room. [2] 7 × 4 = 28 sq m **(b)** Which room is to the south of the Bedroom? [2] The Kitchen (the north arrow shows the top is north, so the room below is south) **(c)** How much larger is the Drawing Room than the Dining room? [2] Dining = 7 × 3 = 21 sq m; 28 − 21 = **7 sq m** **(d)** Find the total floor area of the house. [2] Length 12 m, breadth 4 + 3 = 7 m; 12 × 7 = **84 sq m** (check: 20 + 28 + 15 + 21 = 84)

NUMBER SYSTEM — Marks 9

Q.	Marks	Answer / marking points
5	1	The number 4,56,300 written in the international system is **(b) 456,300**
6	1	Which is the smallest number? **(c) 2,09,867**
7	1	76,482 rounded off to the nearest hundred is **(d) 76,500**
8	6	**(i)** Anita bought 2.35 m of red cloth and 1.8 m of green cloth. (… [3] (a) 2.35 + 1.80 = **4.15 m** (b) 2.35 − 1.80 = **0.55 m** (c) 4.15 × 40 = **₹ 166** **(ii)** The line from 2 to 3 is divided into 10 equal parts. Write t… [3] A = 2.1 B = 2.5 C = 2.9

OPERATIONS — Marks 25

Q.	Marks	Answer / marking points
9	1	The estimated sum of 4,986 and 3,012 (rounding each number to the nearest thousand) is **(a) 8,000**
10	3	Kailash bought 3 kg sugar at ₹ 44 per kg and 2 kg dal at ₹ 126 per kg. He gave a ₹ 500 not… Sugar = 3 × 44 = ₹ 132; dal = 2 × 126 = ₹ 252 Total = 132 + 252 = ₹ 384 Balance = 500 − 384 = **₹ 116**
11	3	Divide 8,205 by 9. Write the quotient and the remainder. Then verify your answer. Quotient = **911**, remainder = **6**. Verification: 9 × 911 + 6 = 8199 + 6 = 8205 = dividend. ✓
12	6	**(i)** For a school picnic, 38 students paid ₹ 125 each and 3 teach… [3] (a) 38 × 125 = ₹ 4,750; 3 × 200 = ₹ 600; total = **₹ 5,350** (b) 5350 − 4500 = **₹ 850** **(ii)** Estimate by rounding (thousands for (a), (b); tens for (c)):… [3] (a) 5,000 + 2,000 = **7,000** (exact 7,011) (b) 9,000 − 4,000 = **5,000** (exact 5,111) (c) 60 × 40 = **2,400** (exact 2,419)

13	6	Bulbs numbered 1 to 30 are fixed in a hall. Switch A lights every 5th bulb (5, 10, 15, …) … **(i)** (a) How many bulbs does Switch A light? (b) How many bulbs d… [3] (a) **6** (5, 10, 15, 20, 25, 30) (b) **3** (10, 20, 30) (c) every multiple of 10 is also a multiple of 5 **(ii)** (a) Write all the factors of 30. (b) Write the common factor… [3] (a) 1, 2, 3, 5, 6, 10, 15, 30 (b) 1, 2, 5, 10 (c) **10**
14	6	**(i)** In the number pyramid each brick is the sum of the two brick… [3] B: 6 + B = 15, so B = **9**. A: 15 + A = 29, so A = **14**. C: 5 + C = 13, so C = **8**. **(ii)** A floor 8 m long and 6 m wide is covered with square tiles o… [3] (a) 8 ÷ 2 = **4** (b) 6 ÷ 2 = **3** (c) 4 × 3 = **12** (check: area 48 ÷ 4 = 12)

MEASUREMENT — Marks 17

Q.	Marks	Answer / marking points
15	3	Fill in the blanks: (a) 4 m 35 cm = _____ cm (b) 2.5 litres = _____ mL (c) 135 minut… (a) 435 (b) 2500 (c) 2 hours 15 minutes
16	3	Write True or False: (a) 10:15 pm is written as 22:15 in the 24-hour clock. (b) 8:30 am is… (a) True (b) False (8:30 am = 08:30) (c) True
17	3	A shop in Alwar opens at 8:30 am and closes at 7:15 pm. For how many hours and minutes is … 7:15 pm = 19:15. 19:15 − 8:30 = 10 h 45 min. (8:30 to 18:30 is 10 h, and 18:30 to 19:15 is 45 min.) Answer: **10 hours 45 minutes**
18	8	The table gives the distance (in km) by road between some cities of Rajasthan. **(a)** What is the distance between Jaipur and Jodhpur? [2] **340 km** **(b)** Which city is nearest to Bikaner among the cities given? [2] **Jodhpur** (250 km; Jaipur is 330 km) **(c)** Sonu goes from Jaipur to Jodhpur through Udaipur. How many k… [2] 400 + 250 = 650 km; 650 − 340 = **310 km** more **(d)** A taxi goes from Kota to Udaipur and returns to Kota. How fa… [2] 2 × 270 = **540 km**

DATA HANDLING AND PATTERNS — Marks 8

Q.	Marks	Answer / marking points
19	1	Which of these is a prime number? **(c) 47**
20	1	A plus-shaped design is made with squares as shown. How many squares will be needed for Fi… **(d) 21 — 1, 5, 9, 13, 17, 21 (add 4 each time)**
21	3	Students of Class 5 were asked to choose their favourite fruit. The tally marks are shown … **(a)** How many students chose Apple? [1] 13 **(b)** Which fruit is the least liked? How many students chose it? [1] Banana, 9 **(c)** How many students were asked in all? [1] 17 + 13 + 9 + 11 = 50
22	3	Observe the pattern: 1 × 9 + 2 = 11 | 12 × 9 + 3 = 111 | 123 × 9 + 4 = 1111. (a) W… (a) 1234 × 9 + 5 (b) 11111 (c) 12345 × 9 + 6 = 111111

END OF ANSWER KEY — Total: 80 Marks